

FAST NUCES

CS 3015 – Robotics Technology

Fall 2024 – Assignment IV

120
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Important Dates

Date Assigned: 25th November 2024

Deadline: 29th November 2024, Friday.

Submission Procedure: Submit to your CR. CR will coordinate with me and submit all the assignments together (once) during the coordinated time.

Late submissions will not be accepted under any circumstances.

Topic	Name	Roll Number	Section	Signature
Member 1	M. Rohaan	221-2327	A	[Signature]
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Formatting Requirements

1. Use A4 white pages with narrow margins.
2. Reports must be neat, clear, and well organized.
3. Include diagrams, figures, and examples to support your explanations.

Group Assignment Guidelines Collaboration Rules

1. This is a group assignment.
2. You are responsible for forming your groups.
3. No excuses will be accepted for incomplete contributions.
4. Each group must have Four (4) members.
5. The assignment will be evaluated as a whole; therefore, the division of work is your responsibility.
6. If your group has fewer than the required number of members, the division of work remains your responsibility.
7. Each group member is expected to contribute actively. Failure to do so may result in penalties or exclusion from the evaluation.
8. Ensure that all group members are familiar with and agree upon the division of tasks.
9. Group disputes will not be entertained. It is the group's responsibility to resolve any internal issues.
10. Group members should peer-review each other's work to maintain a consistent standard of the work's quality and presentation.

Submission Guidelines

1. Fill in the details of group members and attach this 1st page (only) as the title page.
2. The assignment must be handwritten and completed individually.
3. Group submissions should compile all individual reports into a single, organized document.
4. Plagiarized work or solutions copied without understanding will result in a penalty or no credit.
5. Ensure that the final submission includes all necessary components (reports, diagrams, computations, and references).

PULSE WIDTH MODULATION (PWM)

WHAT IS PWM?

Pulse Width Modulation (PWM) is a technique used to control the power delivered to electrical devices. Instead of providing a constant voltage, PWM sends a series of rapid pulses that turn the power 'ON' and 'OFF'. This method allows for efficient control of devices without wasting energy.

Key Concepts

1. Duty Cycle: This is the percentage of time the signal is 'ON' compared to the total cycle time.

For instance, a 50% duty cycle means the signal is 'ON' half of the time.

If the signal is always 'ON', then the signal must have a 100% duty cycle.

The formula to calculate duty cycle is:-

$$\text{Duty Cycle (\%)} = \frac{\text{TURN ON TIME}}{\text{TURN ON TIME} + \text{TURN OFF TIME}} \times 100 \quad \text{OR} = \frac{V_{\text{avg}} \times 100}{V_s}$$

2. FREQUENCY: Measure of the number of pulse cycles that occur in one second.

$$\text{FREQUENCY} = \frac{1}{\text{TIME PERIOD}}$$

How Is A PWM Signal Generated?

A PWM signal is generated using a comparator (a simple electronic device that compares two voltages & checks which of two voltages is higher and sends out a signal based on that.)

1. INPUT SIGNAL: The comparator has two inputs.

i- MODULATING SIGNAL: This is the signal we want to control.
(For eg. a desired speed for a motor)

ii- NON-SINUSOIDAL WAVE OR SAWTOOTH WAVE:

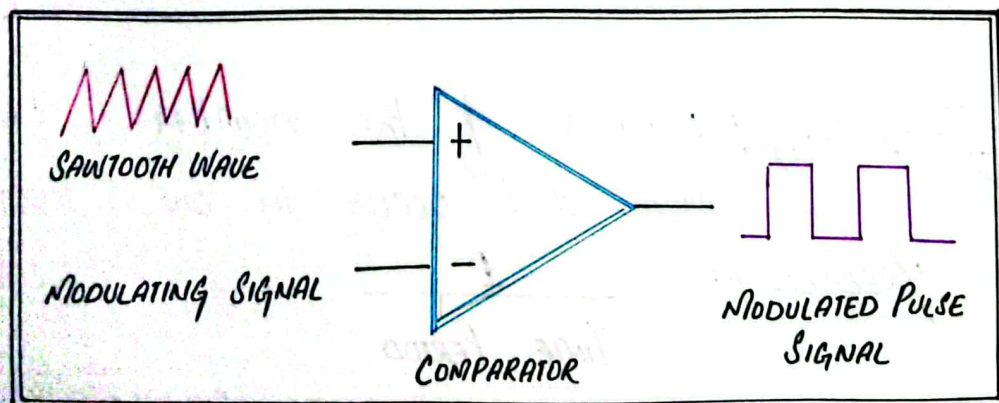
This is a steady wave that varies in a regular pattern.

2. COMPARISON:

The comparator checks these two signals.
If the modulating signal (the one we want to control) is higher than the sawtooth wave, the output is 'ON'.
If it is lower, the output is 'OFF'.

3. OUTPUT PWM SIGNAL:

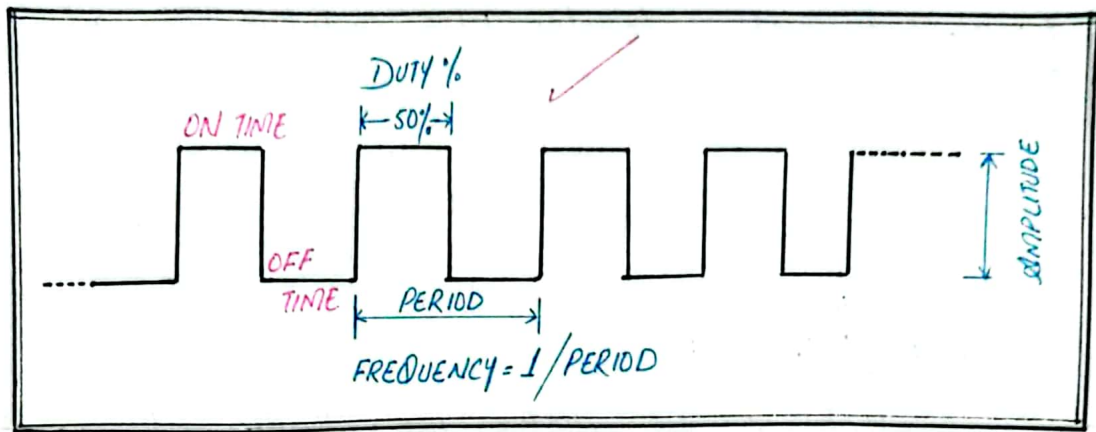
The result is a PWM signal, which consists of a series of pulses that vary in width based on the modulating signal.



HOW IS POWER CONTROLLED?

Power in a PWM signal is controlled by adjusting the 'width of the pulses'.

PULSE WIDTH: The width of each pulse in the PWM signal determines how long the power is 'ON'. A wider pulse means the power is ON for a long time, while narrower means it's for a shorter time.



BENEFITS OF PWM:

1. Efficiency: Reduces heat generation compared to other control methods.
2. Precision: Allows for fine control over power.
3. Cost-Effective: Simplifies circuit design and reduces component costs.

APPLICATIONS OF PWM:

1. Robotics: Used to control servo motors for precise movements.
2. LIGHTING: Dimming LED's in homes and vehicles.
3. POWER SUPPLY: Efficiently converting voltage levels in power supply.

LIMITATIONS:

- 1: Electromagnetic Interference: PWM signals can generate electromagnetic interference and operational noise due to rapid switching.
- 2: Complexity in filtering: Filtering PWM signal to obtain a smooth output can complicate the design and increase costs.
- 3: Limited to certain loads: Not suitable for all types of electrical devices.

NUMERICAL PROBLEM:

Let's control a DC motor using PWM with a duty cycle of 75% at a frequency of 1 kHz. We need to calculate the average voltage supplied if the supply voltage is 12V. Find ON time and OFF time.

GIVEN DATA:

$$\text{Duty Time} = D = 75\%$$

$$\text{Supply Voltage} = V_s = 12V$$

SOLUTION:

We know that,

$$D = \frac{V_{avg}}{V_s}$$

$$V_{avg} = D \times V_s \\ = 0.75 \times 12V = 9V$$

$$\star \text{Average Voltage} = 9V$$

TO FIND:

$$\text{Average Voltage} = ?$$

$$T_{ON} = ? ; T_{OFF} = ?$$

$$T = 1/1000 = 0.001ms$$

$$\text{We know that, } D = \frac{T_{ON}}{T}$$

$$T_{ON} = 0.75 \times 1ms = T$$

$$\star T_{ON} = 0.75ms$$

$$T_{OFF} = T - T_{ON} = 1ms - 0.75ms$$

$$\star T_{OFF} = 0.25ms$$

$$\text{FANIA NALIK 221-2375}$$

MOTORS

WHAT IS A MOTOR?

A motor is a device that converts electricity into motion, usually by spinning.

For Example:

1. In a fan, the motor makes the blades spin to create wind.
2. In a blender, the motor spins the blades to mix or chop food.

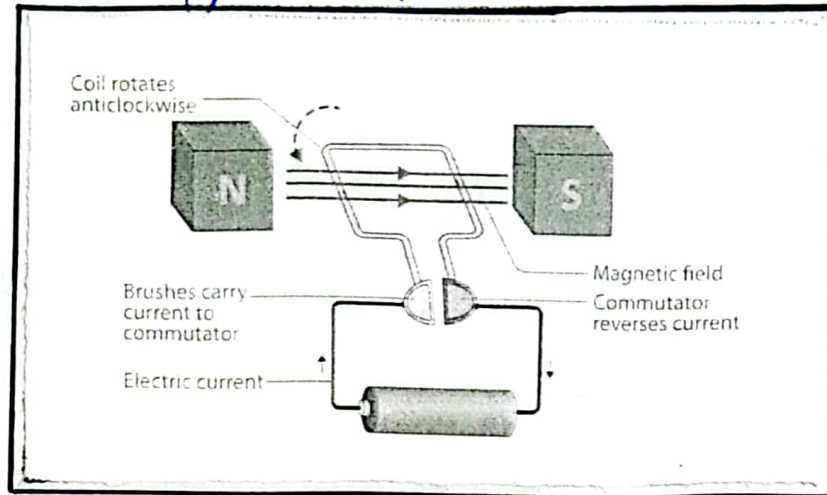
DIFFERENT PARTS OF AN MOTOR AND ITS FUNCTION:

A simple motor has the following parts:

1. POWER SUPPLY: This is like a battery (DC) or plug that gives energy to the motor to make it work.
2. COMMUTATOR: Think of it as a switch that helps the motor change direction to keep spinning smoothly.
3. ARMATURE COIL: This is the spinning part of the motor. It is a long, rectangular coil of wire.
4. ARMATURE CORE: It is the backbone of the motor holding everything in place while supporting the armature coil.

5. FIELD MAGNET: These magnets create a magnetic force that makes the armature coil move.

6. BRUSHES: These are like connectors that let electricity flow b/w moving and non-moving parts



WORKING PRINCIPLE OF A MOTOR:

A motor works because of the interaction b/w electricity and magnets.

1. Magnets Attract and Repel: Magnets have a "push and pull" power. Opposite poles attract and similar poles repel.

2. Electricity Creates a Magnet: When we pass electricity through armature coil, it becomes a temporary magnet.

3. Magnets make it spin: When we place the armature coil b/w the two regular magnets. The magnetic push and pull from the magnets interact with the magnetic force of the wire loop, making it spin.

a motor, this spinning happens continuously because the current keeps flowing, and the magnets keeps interacting. This spinning motion powers fans, toys, etc...

To calculate the force on armature coil, we use the formula:

$$\text{Force} = B \times I \times L \times \sin \theta$$

Where B = Magnetic Field strength

I = Current

L = Length of wire

θ = Angle b/w wire and magnetic field.

APPLICATIONS:

Motors are used in a variety of applications.

1. Drills
2. Industrial Equipment
3. Hard Disk Drives
4. Water pumps

BENEFITS OF MOTORS:

1. Versatility:
2. Automation-Friendly
3. Compact & Lightweight
4. Reliable Performance

LIMITATION OF MOTORS:

1. Heat Generation
2. Noise and Vibration
3. Maintenance Needs
4. Initial Cost: Advance motors can be expensive

NUMERICAL PROBLEM:

A wire of length 1 meter carries a current of 2 amperes and is placed in a magnetic field of strength 0.5 Tesla. The wire is at a 90° angle to the magnetic field. What is the force on the wire?

SOLUTION:

GIVEN DATA:

Length of wires = 1m

Current - $I = 2A$

Magnetic Field Strength = 0.5 Tesla

$\theta = 90^\circ$

We know that

$$F = ILB \sin \theta$$

$$F = (2)(1)(0.5) \sin 90^\circ$$

$$F = 1 \text{ Newton}$$

TO FIND:

Force on wire = ?

Encoder in Robotics

Encoder is an electro-mechanical device that converts mechanical motion to an electrical signal (real time feedback signal) to measure the position, velocity or direction of movement of a rotating shaft or linear motion. Can say that it is the backbone of robotics.

* Encoders are sensors that encode rotation angle (rotary encoder) and linear displacement (linear encoder). Diagram:



Working of Encoder:-

Encoder works by measuring the position/velocity of a rotating shaft (an encoder disk) / linear motion (a coded strip) and generating digital signals to represent this information. The sensing element detects the motion of the shaft and generates signals based on the position of a pattern on the encoder. After processing these signals, the output interface communicates the digital signal to the control system which uses the information to control the motion.

For Examples:-

Joint on a Robotic Manipulator.

Absolute vs Incremental encoders

Absolute encoders can directly provide the current position of the shaft without requiring any previous reference or counting.

Incremental encoders provide a relative position measurement by counting the number of pulses.

Applications for encoders

- * On axis → like a wheeled platform.
- * Navigation → like in delivery robots or automated warehouses.
- * Robotic Arms: Control the movement of arm.
- * Drones & Vehicles → Drones use encoders to track movement and adjust flight paths. Vehicles like Self Driving Cars.
- * 3D Printing: - To build layers of material precisely.

Basic Components of encoders:-

* Sensing Element:-

Detects the motion of the shaft. Most common sensing elements are optical and magnetic sensors.

Optical sensors use light based techniques such as reflection or transmission, while magnetic sensors use magnetic fields to detect changes.

* Encoded disk/Coded Strip:

Encoded disk/Coded strip contains the patterns to determine the position/velocity of the shaft.

* Electronics:-

These are the components that process the signals generated by the sensing elements and convert them to digital.

Benefits:-

- Motor feedback and position measurement all in one
- Improved accuracy
- Speed Control
- Versatility
- Efficiency
- Reliability
- Scalability
- Mobility

Limitations:-

- Cost:- can be expensive
- Resolution:- affects accuracy
- Error Accumulation:- loss of position
- Environmental Sensitivity:-
↳ Sensitive to dirt, dust or moisture.

Numerical Example:-

- Suppose the encoder generates 100 pulses per rotation of the wheel.
- Robot wheel has diameter of 10 cm.

Sol.

$$\text{Circumference, } C = \pi \times d = 3.14 \times 10 = 31.41 \text{ cm}$$

$$\rightarrow \frac{31.41}{100} = 0.3141 \text{ cm per pulse}$$

distance per pulse

Total Distance Traveled

→ Suppose encoder counted 250 pulses,

$$\text{Distance, } S = 250 \times 0.3141 = 78.54 \text{ cm}$$

Conclusion:-

In summary, encoders are essential components in robotics, offering valuable feedback that ensures robots can perform complex tasks with accuracy and efficiency. It helps robots to understand their movement, control their speed and achieve accuracy in tasks. Whether it's a simple robot navigating a room or an industrial robot assembling products, encoders are the eyes that help robots know exactly where they are and what they are doing.

GearBox in Robotics:-

Robotics gearbox are part of joints in robot arms. The joints ensure maneuverability while the gearboxes determine the speed and direction of movement. They are the link between the motor that supplies the drive energy and a machine component that has the task of executing the planned movement.

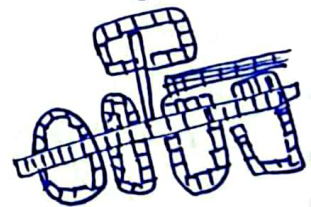
For Example:-

When lifting something heavy, the robot needs more force which the gear box provides.

Working of GearBox:-

The transmission ratio of harmonic drive reducer is larger than the ordinary reducer. Since the number of meshing teeth is large, its carrying capacity and transmission are very high. The harmonic reducer can achieve a space mechanism for passing motion and power such as sealing harmonic transmission deceleration devices and drives spatial mechanisms operating in vacuum, corrosion and other harmful media.

Diagrams:-



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Importance:

- * Motor may rotate too fast for the robot to lift heavy objects
- * Motor might not generate enough torque to handle tasks requiring strength.
- * Gear Box balances speed and force.

Applications of Gear Box:-

- Industrial Robots → Precise control of robotic arms and manipulators.
- Assembly Lines → tasks like welding, material handling with accuracy.
- Heavy Duty tasks → loads and operates in harsh environments
- Medical Robots → Vinci Surgical System
- Consumer Robots → Vacuum Cleaners, drones
- Space Explorations → pick up samples with precise force.

Benefits:-

- Speed & Torque Manipulation
- Flexibility for task-specific requirements
- Protective function for the motor
- Adaptability to Dynamic environments
- Technological Advancements in Gear Box design
- Precision and Energy efficiency

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Limitations of Gear Boxes

- size and Weight → Increase The robot's overall size & weight
- Complexity → More parts means more maintenance
- Energy Loss → Loss of energy as heat due to friction.
- Cost → May expensive
- Torsional Rigidity
- Shape
- Lost Motion
- Load Independent losses
- Maximum Input speed

Types of Gears:-

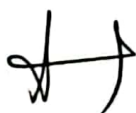
- * Spur Gear
- * Helical Gear
- * Bevel Gears
- * Rack & Pinion Gear
- * Internal Gears
- * Worm Gears

Numerical Example:-

Suppose motor runs at 3000 rotation per minute and produces torque of 1 Nm with a gear ratio of 5:1. speed and torque?

$$\text{New Speed} = \frac{3000}{5} = 600 \text{ Rotation per Minute}$$

$$\text{New Torque} = 1 \times 5 = 5 \text{ Nm}$$

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Conclusion:-

GearBoxes in Robotics enabling precise control over speed and torque. Robots can perform diverse functions - from lifting heavy objects to executing delicate operations with precision.

References for Encoders:-

- * MAB-Robotics
- * Robotics Tomorrow
- * akm.com
- * Indiana University
- * Rozum Robotics

References for GearBox:-

- * GAM Enterprises
- * Igus.eu
- * Research Gate
- * Apexdyna.ni
- * Sagequip.com
- * Robot Park
- * Extruder GearBox Repair

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Interrupts in Robotics:-

Interrupts are signals

that temporarily halt a microcontroller's current process, allowing it to address more urgent tasks. This feature is essential in microcontroller programming for robotics because it enables responsive and efficient handling of multiple inputs ensuring that critical events are managed in real-time.

Interrupts are like an emergency call in a robotic system. When something important happens, an interrupt stops what the robot is doing and makes it pay attention to the urgent task.

For Example:-

If a robot is moving forward and a sensor detects an obstacle, the interrupt tells the robot to stop and take action to avoid a collision.

Types of Interrupts:-

- * External events from hardware peripherals

- * SWI Instruction

Maskable (can be ignored by processor)

Non-Maskable (addressed immediately)

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Importance:-

- Respond immediately to critical situations
- Avoid errors / accidents
- Avoid missing important events.
- React immediately to important events without waiting to finish their current task.

Applications:-

- * Obstacle Detection :- Sensor detects an obstacle
- * Emergency Stops :- Halting all operations
- * Data Communication :- Receiving Real Time data
- * Time-Critical Tasks :- Controlling Motor Speed

Benefits :-

- * Quick Respond / Immediate Respond
- * Efficiency
- * Improved Safety
- * Error Detection & Correction
- * Real-Time processing
- * Resource Saving
- * MultiTasking
- * Energy Efficiency
- * Scalability
- * Reliability
- * Fault Tolerance
- * User Interaction
- * Improve Workflow

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Applications:-

- * Autonomous Vehicle
- * Industrial Automation
- * Home Automation Robots
- * Robotic Arm in Surgery
- * Drones
- * Security
- * Robots
- * Agriculture Robots
- * Educational Robots
- * Warehouse Automation

Limitation of Interrupts:-

- * Complexity
- * Overuse Issues
- * Prioritization Problems
- * Processor Overhead
- * Interrupt Storms
- * Debugging
- * Shared Resources
- * Interrupt Handling
- * Memory Constraints
- * Managing Multiple Sources
- * Real-Time Constraints

Numerical Example:-

Suppose a robot's ultrasonic sensor sends an interrupt signal every 5ms when it detects an obstacle. Robot is moving forward, but it stops when sensor triggered. if $t = 10\text{ms}$ will it respond in time to avoid obstacle.

Sol.

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- * Sensor generates interrupts every 5×10^{-3}
- * Processor checks every 10×10^{-3}
- * The processor will miss every other interrupt because it checks less frequently than the interrupt signal is generated.
- * The robot won't respond in time.

Conclusion:-

Interrupts in robotics enable fast and reliable responses to external events. They make robots efficient, safe and capable of multi-tasking. Proper design is essential to avoid overloading the system or missing critical events.

References:-

- TechTarget
- Robotics Back-End
- Arduino Forum
- Robotics Electronics
- Robot forum
- ResearchGate

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Timers in Robotics:-

Timers enables to precisely time processes, generate signal and count events. A counter converts the number of input cycles to a binary value using an array of triggers. Maximum number of counted cycles depends on the length of this array. A timer in Robotics is a special tool that helps the robots keep track of time. It works like a stopwatch, measuring how much time has passed between events.

For Example:-

Measure the time taken for a robotic arm to move to a certain position or the amount of time a sensor has been activated.

Importance:-

- * Perform Tasks at Specific Intervals :- Like checking a sensor every few seconds.
 - * Control Movement & Speed
 - * Coordinate Multiple Actions
 - * Measure Delays
 - * Timing for feedback loops
 - * Managing Interruptions
 - * Timeout Mechanisms.
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Applications:-

- * **Realtime Monitoring & Diagnostics:-**
↳ like to monitor the temperature of a motor every 10 minutes.
- * **Multi-Robot Coordination:-**
↳ In factory, one robot might be tasked with picking up parts while another might be responsible for packaging.
- * **Autonomous Vehicle Navigation:-**
Uses a timer to check its surroundings at fixed intervals.
- * **Gesture Recognition and Response:-**
Use a timer to determine how long it takes time for a human to raise hand.
- * **Real time Communication in Swarm Robotics:-**
Timers ensure that each robot sends updates to a central system at fixed intervals.

Benefits:-

- * Precision & Accuracy
- * Efficiency
- * Synchronization
- * Autonomy
- * Simplifies Complex Tasks

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- effective
- reduces Computational load
- respond time in real-time Systems
- Facilitates Multitask
- Time-Based motion & speed
- Increases Safety
- Task-Time Limiting
- Predictive Maintenance

Limitations:-

- Dependence on accurate timing
- Resource Consumption
- Fixed Interval Limitations
- Interruptions in Task flow
- Power Consumption
- Overuse in complex systems
- Limited to pre-defined Intervals
- Risk of timing Drift
- Can Cause Resource Contention
- Difficulty in Synchronization

Numerical Example:-

A robot arm needs to lift a box, hold it in place for 5 seconds and then move it to another location. The robot uses a timer to manage the 5 second hold time. If robot move the box at 2:00 PM and $t = 5 \text{ sec}$ then what time will the robot make the box again?

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Sol.

Start time $\rightarrow$ 2:00 PM

$t = 5 \text{ sec}$

timer Activation

After 5 sec, the robot will finish the action and move the box again

$2:00 + 5 \text{ sec} = 2:00:05 \text{ PM}$

Conclusion:-

Timers allowing robots to perform tasks with precision, coordination and efficiency. They enable the robot to manage multiple tasks based on time intervals making them more autonomous and capable. The use of timers requires careful design to avoid issues like resource consumption and errors.

References:-

- * Society of Robots
- * Robotics Lab.eu
- * Robot forum
- * Robotic Lab.eu
- * Micro Robotics
- * RobotShop Community
- * EL3BS1

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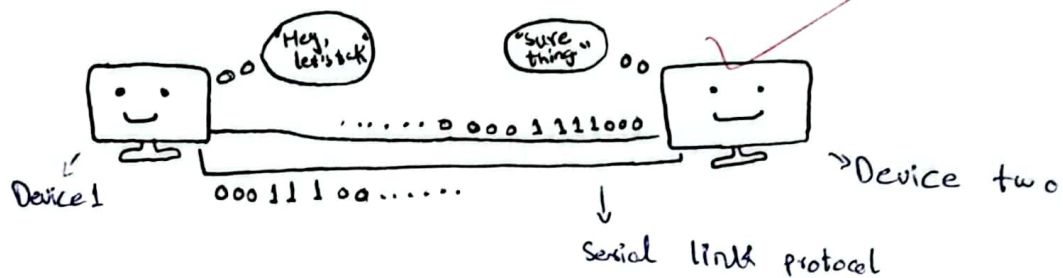
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Serial Link Communication: SPI and I2C Protocols.

Serial communication is a way for electronic devices to "talk". They send 1 bit at a time, so it's almost like only one word is spoken at a time. But it is done really fast due to their fast processing capabilities.



The two most popular methods for this are, SPI and I2C. SPI, refers to "Serial Peripheral interface", and I2C for "Inter-Integrated circuit". Both of these have their Pros and cons,

What is SPI?

SPI is a communication protocol, that connects sensors, displays or memory to a microcontroller. It is like a master-slave relationship. The "master" is usually the device that sends command, i.e., a microcontroller, And "slave", is usually the device that acts on the master's request.

4 wires are used for this communication, namely,

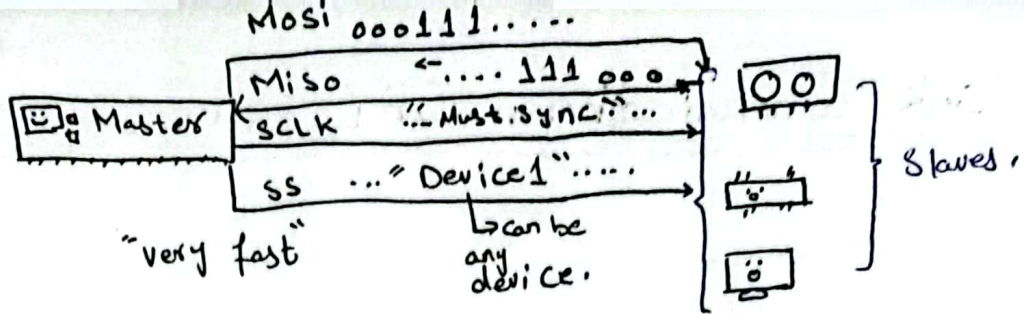
MOSI, Sends data from master to slave

MISO, Sends data from slave to master

SCLK, A clock line that ensures sync

SS, Selects the device master is talking to,

2



SPI, is usually really fast, i.e, if you wanna control a display or read real-time sensor data, SPI will ensure there is no delay. But, since it requires a lot of wires, it may not be ideal for many devices.

For example, imagine a sensor sends temperature data to a microcontroller using SPI, if the SPI clock speed is 8MHz, it can transfer 1kB of data in just 1 millisecond. That is like flash.

So,

Advantages,

- 1) very fast and reliable for large data transfer.
- 2) can send and receive data at same time.

Disadvantages,

- 1) Requires a lot of wires comparatively.
- 2) Not ideal for many devices.

is I2C?

I2C, is another way for devices to communicate. It is simpler than SPI and uses only two wires, namely,

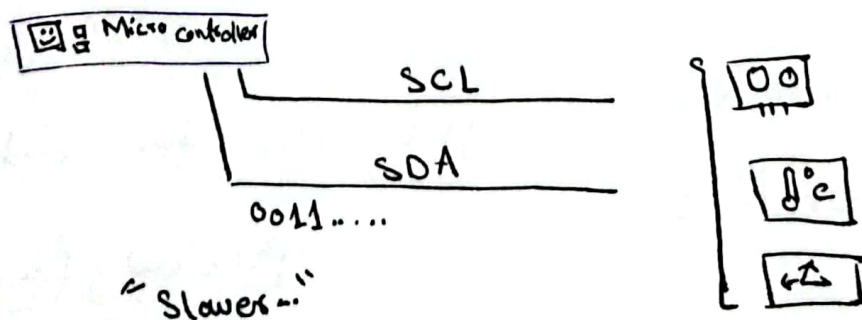
SCL, The clock line.

SDA, The data line.

This allows many devices to easily and simply connect. Each device has its address so it's easy for the microcontroller to know whom to talk to. It is like a shared conversation where everyone takes turns to speak.

Although, comparatively it is slower than SPI, so it's better to be used with devices not requiring fast data transfer.

For example, consider 256 bytes of data is transferred using I2C at 100 kHz clock speed. I2C also adds an extra bit for acknowledgment, so it would take 23 milliseconds, which is slower than SPI, but does the job.



④ So,
Advantages,

- 1) fewer lines, so alot simpler.
- 2) Connects Many devices using only 2 wires.

Disadvantages,

- 1) Slower and not suitable for high speed tasks
- 2) Tricky since shared communication is used.

Conclusion,

Both have their pros and cons, so knowing when to use which would truly harness their abilities.

References,

- Arduino documentation for SPI and I2C.
- "Embedded systems" by Jack Ganssle.
- Texas instruments technical notes on communication protocols.
- OpenAI's gpt 4o for web search
- OpenAI's gpt o1-preview for verification.